

Text-to-Speech (TTS) Project Documentation

Overview

This project converts a given text into speech using Python's pyttsx3 library. The speech is saved as an audio file, and it can be played back for listening. The project allows customization of the voice (male or female) and adjusts the speed and volume of the speech.

Key Features

- Text-to-Speech Conversion: The project reads out a block of text using a chosen voice.
- Voice Selection: It allows switching between male and female voices.
- Audio Customization: The speed (rate) and volume of the speech can be adjusted.
- Save Audio: The speech is saved as an .mp3 file that can be played back.
- Playback: The saved audio can be played using the IPython Audio function.

Steps

1. Initialization of pyttsx3:

- The pyttsx3 library is initialized to convert text into speech.

2. Voice Configuration:

- The available voices (male and female) are fetched using `audio.getProperty('voices')`.
- The voice is set to male (or female) based on the choice using `audio.setProperty('voice', voice[0].id)` for male or `voice[1].id` for female.

3. Text-to-Speech Conversion:

- The provided text is converted into speech using the `audio.say(text)` function.

4. Saving the Audio:

- The speech is saved to an `.mp3` file using `audio.save_to_file(text, 'test_male_Voice.mp3')`.

5. Playing the Audio:

- The saved audio file is played back using the `IPython Audio()` function.

Output

- An audio file (`test_male_Voice.mp3`) is created containing the speech of the provided text.
- The user can play the audio file within an IPython environment.

Use Cases

- Accessibility: Helps in converting text to speech for visually impaired users.
- Voice Applications: Useful in building voice-based assistants or applications.
- Learning Tools: Can be used to create audio content for e-learning or language learning apps.

Conclusion

This project demonstrates a simple way to convert text into speech and save it as an audio file. It provides flexibility in voice selection and allows for easy customization of speech rate and volume, making it ideal for various applications.